Research Proposal: Marking Rubric and Feedback Form

Foundations of Clinical Research · Level 1: From Clinical Question to Publication

Organisation: Neudata (*#ClearDataClearImpact*)

Total: 100 points for everyone, worked out in one of two ways:

- **Presenters** (the about 5 volunteers who present in Session 8): written proposal and publication plan 70 + presentation 30 = 100.
- **Non-presenters** (everyone else): written proposal and publication plan only, scaled to 100: **total = written subtotal (out of 70) × 100 / 70**, rounded to the nearest whole point. Example: 49 / 70 gives 70 / 100.

The slides that non-presenters submit are not scored; the instructor reads them only to check that they match the written proposal. Always report the written subtotal (out of 70) as well, so that everyone can be compared on the same written work.

Use with: `Proposal_Brief.pdf` and `Presentation_Guide.pdf`

> **The marks are formative. They help participants improve their proposal and help instructors see which parts of the course need strengthening. Presenting is voluntary and never lowers a score. Reward clear thinking and honesty about limitations over polish. A simple, feasible study planned well scores higher than an ambitious study planned vaguely.**

How to score

1. Read the whole proposal once without scoring.
2. For each criterion, choose the level that best describes the work and give its points. If the work sits between two levels, choose the lower one and note what would lift it.
3. Write **one comment per criterion that loses points**, saying what to add or change. These comments are the most valuable part of the feedback.
4. Add the points. Use the bands at the end only as a guide for the overall comment.

Levels: Excellent: complete, correct and specific. **Good**: mostly complete and correct, minor gaps. **Needs work**: present but vague, incomplete or partly incorrect. **Missing**: absent or wholly incorrect.

A. Written proposal and publication plan (70 points)

Part 1: Sessions 1 and 2 (Ask; Find and read)

Criterion	Excellent (6)	Good (4)	Needs work (2)	Missing (0)
1. Problem statement, rationale & working title	Clear clinical problem, sized with a number and a source; states what is not known and why it matters locally; informative working title built from the PICO.	Problem and importance clear; gap or size of the problem only implied.	Problem described in general terms; no evidence of size or gap.	No problem statement.
2. PICO, research question & FINER	All PICO elements specific; question follows directly from the PICO; FINER check is honest and changed something.	PICO complete but one element vague (e.g. outcome without time point); FINER present.	PICO incomplete or question does not match it; FINER is a list of "yes".	No PICO or no question.
3. Aim, objectives, hypothesis & references	One aim, 2–3 measurable objectives, a testable hypothesis (where relevant); 5 relevant, correctly cited references.	Objectives present but one not measurable; 3–4 good references.	Objectives restate the aim; hypothesis not testable; references few, old or not relevant.	Absent.

Part 2: Sessions 3 and 4 (Design; Bias and effect)

Criterion	Excellent (6)	Good (4)	Needs work (2)	Missing (0)
4. Design & justification	Design correctly named and well matched to the question; explains why a stronger design is or is not possible.	Appropriate design, brief justification.	Design misnamed or poorly matched to the question; no justification.	No design stated.
5. Setting, population, exposure & outcomes	Reproducible inclusion/exclusion criteria; exposure and comparator defined; one primary outcome with how and when it is measured; sensible secondary outcomes.	Mostly clear; one element vague (e.g. outcome timing).	Criteria vague; several primary outcomes or none defined.	Absent.
6. Bias & how it is reduced	At least two specific threats relevant to *this* study (e.g. confounding by indication) with a concrete step for each.	Threats named with general remedies.	Generic list of biases not linked to the study.	No discussion of bias.

Part 3: Sessions 5 and 6 (Data; Analyse)

Criterion	Excellent (6)	Good (4)	Needs work (2)	Missing (0)
7. Variables, mini-CRF, data dictionary & data management	Variables table with correct types and units; 6–10 clear CRF items consistent with the dictionary; data capture, checks, secure storage and audit trail described concretely.	Minor errors in types or units; CRF mostly clear; general data-quality plan.	CRF items ambiguous (no units or time windows); dictionary missing or inconsistent; no data-management plan.	Absent.
8. Sample size, analysis & missing-data plan	Sample size statement with assumptions and source, or a precision target for descriptive studies; summaries and test/model match the variable types; named confounders; one pre-specified primary outcome; sensible missing-data plan (prevention, reporting, handling).	Sample size given with some assumptions; analysis mostly appropriate; missing data mentioned but plan thin.	A number without assumptions; test does not match the outcome type; no missing-data plan.	Absent.

Part 4: Session 7 (Protect and write)

Criterion	Excellent (6)	Good (4)	Needs work (2)	Missing (0)
9. Ethics, registration, confidentiality & feasibility	Ethics route and appropriate consent (including patients lacking capacity where relevant); registration (where and when) or why not needed; de-identification and storage; realistic timeline (Gantt), budget headings and recruitment justified by unit numbers.	Most elements present; one or two thin (e.g. recruitment not justified).	Ethics mentioned only as "will apply for approval"; no registration; unrealistic timeline.	Absent.
10. Publication plan	Suitable, checked target journal (+ back-up); correct reporting guideline for the design; authors meet ICMJE criteria with CRediT roles; refined working title; structured-abstract skeleton with placeholders (no invented results) and calibrated conclusions; local dissemination.	Journal, guideline and authors present; abstract skeleton incomplete or roles vague.	Wrong reporting guideline, unchecked journal, gift authorship, or abstract with invented results or causal claims.	Absent.

Whole document

Criterion	Excellent (10)	Good (7)	Needs work (3)	Missing (0)
11. Coherence & clarity	Parts agree with each other (same question, outcome, population and title throughout); plain language; within about 6–8 pages; template used; references managed consistently; AI use statement if relevant.	Minor inconsistencies or some wordiness.	Parts contradict each other (e.g. different primary outcomes); hard to follow.	Not a usable document.

Written proposal and publication plan subtotal: _____ / 70

B. Presentation in Session 8 (30 points, presenters only)

Score the 6-minute proposal presentation of the volunteers only. Skip this section for non-presenters and use the scaling rule above.

Criterion	Excellent (6)	Good (4)	Needs work (2)	Missing (0)
12. Structure & content	All 6 slides as in the guide; the key decisions (question, design, outcome, sample size, ethics, journal and reporting guideline) are visible.	One slide thin or one key decision missing.	Several parts missing or out of order.	Did not present.
13. Clarity & plain language	Message understandable to every health professional in the class; one message per slide.	Mostly clear; some jargon or crowded slides.	Hard to follow; slides read aloud.	Not applicable.
14. Visual design	Readable on a shared screen (laptop size); pictures, tables or diagrams instead of text blocks (PICO table, flow, timeline).	Readable; mostly text.	Small fonts, walls of text.	Not applicable.
15. Timekeeping	Finished between 5:00 and 6:00.	Finished by 6:30.	Stopped by the chair at 6:30, or under 4 minutes.	Not applicable.
16. Questions & discussion	Answers questions directly and honestly; acknowledges limitations; responds to the chat.	Answers adequately; some questions only partly answered.	Defensive or unable to answer basic questions about the design.	Not applicable.

Presentation subtotal: _____ / 30

Total, presenters: written _____ / 70 + presentation _____ / 30 = _____ / 100

Total, non-presenters: written _____ / 70 × 100 / 70 = _____ / 100

Guide to overall comment

Total	Overall comment
80 to 100	Strong proposal, ready to develop into a full protocol with the research office.
60 to 79	Good proposal. Revise the flagged sections before taking it further.
40 to 59	Promising idea, but the question or design needs rethinking before it can go further.
Below 40	Major revision needed. Meet an instructor to agree the next steps.

Common problems to look for (and comment on)

- Outcome without a time point ("mortality" instead of "28-day mortality").
- An RCT proposed where randomisation is unethical or impossible, or a cohort where a simple RCT would be feasible.
- "We will adjust for confounders" without naming them.
- Sample size given as a number with no assumptions, or copied from another study.
- Recruitment that the unit cannot deliver (check the admissions per month).
- "Consent will be obtained" for unconscious ICU patients without saying from whom.
- CONSORT named for an observational study, or STROBE for a trial.
- A journal chosen from an email invitation, or no check of scope and fees.
- No registration planned for a trial, or registration "after the results".
- Missing data ignored, or "we will exclude patients with missing data" as the only plan.
- An abstract skeleton with invented numbers, or a conclusion that claims causation from an observational design.
- Authors listed for seniority (gift authorship), or CRediT roles missing.

Feedback form: proposal presentation (one per presenter)

Presenter: _____ **Reviewer:** ☐ Instructor ☐ Co-instructor **Date:** _____

Time taken: _____ min ☐ 4-minute ☐ 5-minute ☐ 6-minute time signal reached

Slide by slide (tick one box per slide)

Slide	Clear	Partly clear	Unclear / missing
1. Title & presenter	☐	☐	☐
2. Problem & PICO	☐	☐	☐
3. Design & participants	☐	☐	☐
4. Data & analysis plan	☐	☐	☐
5. Ethics & feasibility	☐	☐	☐
6. Publication plan & impact	☐	☐	☐

Star 1 (something specific that was strong):

_____

Star 2 (something else that was strong):

_____

Wish (the one change that would most improve the written proposal):

_____

_____

One question we would ask the presenter:

_____

Presentation score, instructor only (criteria 12 to 16): ___ + ___ + ___ + ___ + ___ = _____ / 30